MILITARY SPECIFICATION

POWER CONVERTER FOR AN/VSG-2

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Power Converter for the Sight, Thermal, Tank Mounted, AN/VSG-2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, and Seals
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-B-117	- Bags, Sleeves and Tubing-Interior Packaging
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible for Packaging

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners, or Palletized Type Base
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment
MS-24535	- Fastener-Container Closure

DRAWINGS

DLSM-B-804914	- Power Converter, AN/VSG-2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition. The Power Converter, AN/VSG-2, herein known as the converter, converts tank power to the voltages necessary for operation of the AN/VSG-2. The power converter is connected to the tank power source by cable 1W5, which is supplied by the tank contractor, and the remaining AN/VSG-2 components by the interconnect cable assembly 3W4 supplied with the AN/VSG-2. The power converter contains electronic circuits and protective devices necessary to operate the system over the entire environmental and electrical environment of the tank.

3.2 Construction. The converter shall be constructed in accordance with DLSM-B-804914 and as specified herein.

3.3 First article. When specified in the contract (see 6.2), the contractor shall furnish 5 converters for first article inspection and approval.

3.4 Parts, materials and processes. Parts, materials and processes shall be as specified herein and as shown on the applicable drawings. Parts, materials and processes not specified shall be selected in accordance with MIL-P-11268.

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3.5 Major components. The converter consists of the following major components and subassemblies:

a. Housing assembly
b. Relay assembly
c. Transformer assembly
d. Driver heatsink assembly
e. Regulator heatsink assembly
f. Controller PWB
g. Inverter PWB
h. Video rectifier filter PWB
i. Auxiliary rectifier filter PWB
j. Video regulator PWB
k. Auxiliary regulator PWB

3.6 Preconditioning. Each converter shall be subjected to a 48-hour preconditioning prior to delivery.

3.7 Operation.

3.7.1 Input power characteristic. When loaded per 3.7.7, the input power shall not exceed 300 watts with input voltage anywhere between 18 volts direct current (Vdc) and 30 Vdc.

3.7.2 Reverse voltage protection. With input voltage between minus 18 Vdc and minus 30 Vdc, the circuit breaker shall interrupt the input power.

3.7.3 Overvoltage protection. With input voltage at 36 Vdc, the circuit breaker shall interrupt the input power.

3.7.4 400 Hz output. With the 120 volts alternating current (Vac), 400 Hz output loaded to 64 watts, the voltage shall be 120, (+10 -12) Vac, 400, ±20 Hz.

3.7.5 Regulated dc voltages. Eight output voltages shall be present when the current is at maximum as follows:

Output voltage (Vdc)	Max continuous current (A)
+10.5 ± 0.5	1.0
+10.0 ± 0.5	1.0
-9.0 ± 0.5	1.0
+10.0 ± 0.8 -0.5 (LED)	2.0
+5.0 ± 0.3	0.750
-5.0 ± 0.3	0.500
+15.0 ± 0.5	1.0
-15.0 ± 0.5	0.750

3.7.6 Overcurrent/short circuit. The 8 specified voltages of 3.7.5 shall drop below 5 Vdc when a suitable load is applied to each and shall return to normal when the load is removed (see 3.7.7). A short circuit of the 120 Vac, 400 Hz output shall trip the circuit breaker. Afterwards, when loaded to 64 watts with the circuit breaker reset, the voltage shall conform to 3.7.4.

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3.7.7 Low input voltage. With 18 Vdc supplied as an input, the 120 Vac, 400 Hz voltage loaded to 64 watts shall conform to 3.7.4. The regulated dc voltage when loaded shall be as follows:

DC voltages (Vdc)	Voltage ripple (mV)	Load current (A)
+10.5, ±0.5	25	0.8
+10.0, ±0.5	25	0.8
-9.0, ±0.5	25	0.8
+10.0, +0.8 -0.5 (LED)	25	1.6
+5.0, ±0.3	50	0.6
-5.0, ±0.3	50	0.5
+15.0, ±0.5	50	0.8
-15.0, ±0.5	50	0.6
AC voltage		Load power
120 Vac, 400 Hz		64 watts

3.7.8 Sealing. The converter shall show no evidence of leakage in excess of 0.2 psig when subjected to an internal pressure of 5, ±1 psig for a minimum of 2 hours. The interior atmosphere of the units shall contain dry nitrogen (having a maximum dew point of -32°C at atmospheric pressure), at a pressure of 0.25 to 5.0 psig when delivered.

3.7.9 Transient power. The AN/VSG-2 shall operate normally during the transient voltage defined in figure 1 without loss of imaging capability. The AN/VSG-2 shall not be damaged by the transient voltages defined in figure 2, following reset of the circuit breakers.

3.8 Environment.

3.8.1 Low temperature.

3.8.1.1 Operation. The converter shall not be damaged (see 6.3) when operated at temperatures to -32°C.

3.8.1.2 Storage. The converter shall not be damaged when subjected to storage temperatures to -57°C.

3.8.2 High temperature.

3.8.2.1. Operation. The converter shall not be damaged when operated at temperatures to +52°C.

3.8.2.2 Storage. The converter shall not be damaged when stored at temperatures to +71°C.

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3.8.3 Temperature shock. The converter shall not be damaged when subjected to sudden changes in temperature between -57°C and +71°C following a 4-hour soak at each temperature.

3.8.4 Altitude.

3.8.4.1 Operation. The converter shall not be damaged when operated at altitudes up to 10,000 feet above sea level.

3.8.4.2 Transportation. The converter shall not be damaged when subjected to altitudes up to 40,000 feet above sea level.

3.8.5 Shock.

3.8.5.1 Basic shock. The converter shall not be damaged when subjected to basic shock of 35 g's at 15 milliseconds duration.

3.8.5.2 High intensity shock. The converter shall not be damaged when subjected to high intensity shock of 80 g's at 2 milliseconds duration (group C: 60 g's at 2 milliseconds).

3.8.5.3 Bench handling shock. The converter shall not be damaged when subjected to shocks encountered in bench handling.

3.8.6 Vibration. The converter shall not be damaged when subjected to the following random and superimposed sinusoidal levels for a period of 15 minutes in each of three mutually perpendicular axes (group C: 5 minutes per axis).

Random	2×10^{-3} g^2/Hz, 5 to 500 Hz
Sinusoidal	0.11 g's, X axis 0.67 g's, Y axis 0.34 g's, Z axis Octave 1 42 to 84 Hz Octave 2 84 to 168 Hz

3.8.7 Waterproofness. The converter shall withstand water hosing at a minimum of 5 pounds per square inch (psig) through a 1-inch nozzle parallel to the joints from a distance of 3 feet maximum for a time of 5 minutes minimum, without intake of water or damage.

3.8.8 Humidity. The converter shall not be damaged by exposure to 94, +4 percent relative humidity.

3.8.9 Dust. The converter shall not be damaged when exposed to dust at velocities up to 1750, ±250 feet per minute (fpm) for 12 hours.

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3.9 Reliability. The converter shall have a minimum acceptable mean time between failure (MTBF) of 18,181 hours.

3.10 Workmanship. Workmanship shall be as specified in MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Materials inspection. The supplier is responsible for processes, parts and materials used in manufacturing the converter in accordance with referenced specifications, standards and drawings as specified herein.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (4.3)
b. Quality conformance inspection (4.4)
c. Inspection of packaging (4.10)

4.3 First article inspection. When specified in the contract or purchase order, the contractor shall furnish 5 converters for first article inspection (see 6.2).

4.3.1 Inspection. Each assembled first article converter shall be examined in accordance with table I. Presence of one or more defects shall be cause for rejection of the converter.

4.3.2 Preconditioning. Each converter which has passed the inspection of 4.3.1 shall be subjected to the preconditioning of 4.6.

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TABLE I. Inspection.

Defects	Requirement paragraph	Inspection criteria
101. Any part or component missing or damaged	3.2	MIL-STD-252
102. Finishes not protecting substrates (scratches, etc)	3.2	MIL-STD-252
103. Improper or absent marking or labels	3.2	MIL-STD-252
104. Color of finishes incorrect	3.2	MIL-STD-252
105. Workmanship not as specified	3.10	MIL-STD-454

4.3.3 Tests.

4.3.3.1 First article inspection. From converters that have passed preconditioning of 4.3.2, 5 first article converters shall be subjected to all the tests of table II, in any order. Unless otherwise specified, the tests of table II shall be performed with the converter at the ambient conditions (see 6.3). Failure of any converter to pass all tests in table II shall constitute failure of first article.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Input power characteristic	3.7.1	4.7.1
Reverse voltage protection	3.7.2	4.7.2
Overvoltage protection	3.7.3	4.7.3
400 Hz output	3.7.4	4.7.4
Regulated dc voltages	3.7.5	4.7.5
Overcurrent/short circuit	3.7.6	4.7.6
Low input voltage	3.7.7	4.7.7
Sealing	3.7.8	4.7.8
Transient power	3.7.9	4.7.9
Low temperature	3.8.1	4.8.1
High temperature	3.8.2	4.8.2
Temperature shock	3.8.3	4.8.3
Altitude	3.8.4	4.8.4
Shock	3.8.5	4.8.5
Vibration	3.8.6	4.8.6
Waterproofness	3.8.7	4.8.7
Humidity	3.8.8	4.8.8
Dust	3.8.9	4.8.9
Reliability	3.9	4.9

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4.4 Quality conformance inspection.

4.4.1 Inspection. Each assembled converter shall be examined in accordance with table I. Presence of one or more defects shall constitute failure of that converter.

4.4.2 Preconditioning. Each converter which has passed the inspection of 4.4.1 shall be subjected to the preconditioning of 4.6.

4.4.3 Group A inspection. Each converter that has passed the preconditioning of 4.4.2 shall be inspected for the attributes specified in table III, in any order. Failure of any converter to pass all the tests of table III shall constitute failure of that converter.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Reverse voltage protection	3.7.2	4.7.2
Overvoltage protection	3.7.3	4.7.3
400 Hz output	3.7.4	4.7.4
Regulated dc voltages	3.7.5	4.7.5
Overcurrent/short circuit	3.7.6	4.7.6
Low input voltage	3.7.7	4.7.7
Sealing	3.7.8	4.7.8

4.4.4 Group B inspection. Group B inspection is not applicable.

4.4.5 Group C inspection. Inspection lots (see 6.2) shall be formed from converters which have passed the group A inspection of 4.4.3.

4.4.5.1 Monthly group C inspection. Five converters shall be randomly selected from each lot and inspected for the attributes specified in table IV. Failure of any converter to pass all the tests of table IV shall constitute group C failure.

TABLE IV. Group C monthly inspection.

Inspection	Requirement paragraph	Test paragraph
Input power characteristic	3.7.1	4.7.1

4.4.5.2 Trimonthly group C. Two converters which have passed the tests of 4.4.3 and 4.4.5.1 shall be randomly selected from the first lot and every third lot thereafter and inspected for the attributes specified in table V. Failure of any converter to pass all the tests of table V shall constitute group C failure.

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TABLE V. Group C trimonthly inspection.

Inspection	Units	Requirement paragraph	Test paragraph
Low temperature	2	3.8.1	4.8.1
High temperature	2	3.8.2	4.8.2
Temperature shock	2	3.8.3	4.8.3
Altitude	2	3.8.4	4.8.4
Shock	1	3.8.5	4.8.5
Vibration	1	3.8.6	4.8.6
Waterproofness	1	3.8.7	4.8.7
Humidity	1	3.8.8	4.8.8

4.4.5.3 Group C failure criteria. Actions relative to group C failure shall be as specified in the contract or order (see 6.2).

4.4.5.4 Reinspection of conforming group C. Converters which have been subjected to and passed group C inspection of 4.4.5.1 and 4.4.5.2 shall have all damage (see 6.3) repaired and be reinspected, per 4.4.3. Those converters which subsequently pass all tests of 4.4.3 shall be accepted on contract or order.

4.5 Group D inspection. Group D inspection is the production reliability test of 4.9. A minimum of three samples shall be randomly selected to perform the group D inspection. The samples shall have previously passed group A inspection. The group D inspection shall be initiated three months after first article reliability completion and repeated once more so as to be completed three months prior to completion of production.

4.6 Preconditioning. Each converter shall be subjected to a minimum of 6 preconditioning cycles. Each cycle shall conform to the profile of figure 3. The temperature extremes shall be $-32^{\circ}C$ and $+52^{\circ}C$. The system shall be vibrated at 2.2 g's at a nonresonant frequency between 20 Hz and 60 Hz for 10 minutes each operating hour. If damage occurs, the system shall be removed from this test, repaired and returned to preconditioning. The preconditioning shall continue for at least one full cycle without additional failure after repair.

4.7 Tests. All tolerances herein are absolute. Any test measuring equipment tolerances shall be subtracted from the converter assembly tolerance. Unless otherwise specified, no internal adjustments or component replacement shall be made on any converter subsequent to the start of any of the tests described herein.

4.7.1 Input power characteristic. The converter shall be used to measure the input voltage at the input connector. A dc ammeter shall be connected in series with the input line. The 120 Vac, 400 Hz output shall be loaded with

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a 64 watt load. The 8 regulated dc outputs shall be loaded as indicated in 3.7.7. Set the input voltage for 30 Vdc and measure the current. The product of the input voltage and the current shall not exceed 300 watts. The converter shall be turned off and then turned back on with the input voltage set for 18 Vdc. The current shall be measured. The product of input voltage and current at 30 Vdc and at 18 Vdc in excess of 300 watts, failure to start, or failure to operate per 3.7.7, shall constitute failure of this test.

4.7.2 Reverse voltage protection. Connect the converter input to -18 Vdc and -30 Vdc power source. The circuit breaker on the converter input shall interrupt the input power. Failure to interrupt the input power shall constitute failure of this test.

4.7.3 Overvoltage protection. Connect the converter input to a 36 Vdc power source. The circuit breaker on the converter shall interrupt the input power. Failure to interrupt the input power shall constitute failure of this test.

4.7.4 400 Hz output. With an input of 30 Vdc connect a 64 watt load to the 120 Vac, 400 Hz output. Observe the 120 Vac, 400 Hz output with an oscilloscope. Failure to display a 400 Hz square wave in accordance with 3.7.4 shall constitute failure of this test.

4.7.5 Regulated dc voltages. Connect a dc ammeter in series with the outputs of the power converter and measure the outputs with a dc voltmeter. Adjust the loads for each output for the maximum current as listed in 3.7.5. Failure to meet the requirements of 3.7.5 shall constitute failure of this test.

4.7.6 Overcurrent/short circuit. Apply a suitable load to each regulated dc output, one at a time, and observe that the voltage drops below 5 volts. Reapply normal loads per 3.7.7 and observe that the voltage returns to normal. Short circuit the 120 Vac output and observe that the circuit breaker trips. Remove the short circuit, reset the circuit breaker and load the output to 64 watts and observe that 120 Vac returns to normal per 3.7.4. Failure of each dc output to return to normal, failure of the circuit breaker to trip when the 120 Vac output is short circuited or failure of the 120 Vac to return to normal per 3.7.4, after resetting the circuit breaker, shall constitute failure of this test.

4.7.7 Low input voltage. With the input voltage at 18 Vdc, the regulated voltages and 120 Vac shall be measured. Failure to comply with 3.7.7 shall constitute failure of this test.

4.7.8 Sealing. Immediately prior to the start of the sealing demonstration, all access covers, plates, etc, shall be removed and installed 3 times and all controls exercised through their full range 3 times. This test shall be performed utilizing a pressure gage capable of reading between 0.25 and 6.0 psig

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with a resolution of 0.2 psig or less. Pressurize the converter with dry nitrogen gas having a dew point at least as low as -32°C to 5.0, ±1.0 psig. Retain the applied pressure in the converter for a minimum of 2 hours. At the end of the 2-hour period, there shall be no evidence of leakage in excess of 0.2 psig as indicated on the pressure gage to determine compliance with 3.7.8. Upon satisfactory completion of the sealing test, the converter shall be flushed with dry nitrogen gas and sealed with delivery pressure as specified in 3.7.8. Failure to meet the requirements of 3.7.8 shall constitute failure of this test.

4.7.9 Transient power. The AN/VSG-2 operating and experiencing the transient voltages defined in figure 1, shall exhibit no raster movement, display flashing, electronic noise on the display or blooming that would cause loss of image to an observer. The AN/VSG-2 operating and experiencing the transient voltages defined in figure 2 shall withstand the input by presenting a normal image immediately after the transient has stabilized to the steady state voltage (25 Vdc) of figure 2. Raster movement, display flashing, electronic noise visible on the display, blooming for periods longer than 50 milliseconds while experiencing the transient input of figure 1, or inability to survive without damage the transient input of figure 2 shall constitute failure of this test. The circuit breaker may be reset, if necessary.

4.8 Environmental tests. Unless otherwise specified, environmental tests shall be performed in accordance with MIL-STD-810, except when modified by this specification. The performance tests required before, during and after the completion of the individual tests of MIL-STD-810 shall be 4.7.4 and 4.7.5.

4.8.1 Low temperature. The converter shall be tested in accordance with MIL-STD-810, Method 502, Procedure I, except that the storage period (Step 2) shall be 12 hours at -57°C, and the operating temperature (Step 4) shall be -32°C. Failure to meet the requirements of 3.8.1 shall constitute failure of this test.

4.8.2 High temperature. The converter shall be tested in accordance with MIL-STD-810, Method 501, Procedure II. The operating temperature (Step 7) shall be +52°C. Failure to meet the requirements of 3.8.2 shall constitute failure of this test.

4.8.3 Temperature shock. The converter shall be tested in accordance with MIL-STD-810, Method 503, Procedure I. Failure to meet the requirements of 3.8.3 shall constitute failure of this test.

4.8.4 Altitude. The converter shall be tested in accordance with MIL-STD-810, Method 500, Procedure I, except that the chamber pressure, operating, shall be 523.0 mm Hg or 10,000 feet above sea level. The altitude, nonoperating, shall be 40,000 feet, shall be maintained for a period of not less than one hour, and shall be performed prior to the operating altitude test. Failure to meet the requirements of 3.8.4 shall constitute failure of this test.

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4.8.5 Shock. The converter shall be tested in accordance with MIL-STD-810, Method 516.1, Procedures I, IV and V. Figure 516.1-2 shall apply to Procedures I and IV. The parameters for Procedure I shall be 35 g's peak value with 15 ms duration. The parameters for Procedure IV shall be 80 g's peak value with 2 ms duration (group C: 60 g's at 2 milliseconds) applied by 3 shocks along each axis (18 shocks total). Failure to meet the requirements of 3.8.5 shall constitute failure of this test.

4.8.6 Vibration. The converter shall be tested in accordance with MIL-STD-810, Method 514.1, as modified herein. Mount the converter on the vibration table. While non-operating, the converter shall be vibrated in each of 3 mutually perpendicular axes at input levels of 3.8.6 for 15 minutes (group C: 5 minutes per axis). The appropriate periodic component will be swept up each octave in one minute and down the octave the next minute. Failure to meet the requirements of 3.8.6 shall constitute failure of this test.

4.8.7 Waterproofness. Subject the converter to a water pressure of 5 psig through a 1-inch nozzle from a distance of 3 feet maximum in a direction parallel to the joints for 5 minutes. Wipe the converter dry, remove the covers and inspect for water inside. Any evidence of water inside shall constitute failure of this test.

4.8.8 Humidity. The test shall be conducted in accordance with MIL-STD-810, Method 507, Procedure II. Failure to meet the requirements of 3.8.8 shall constitute failure of this test.

4.8.9 Dust. The test shall be conducted in accordance with MIL-STD-810, Method 510, Procedure I. Failure to meet the requirements of 3.8.9 shall constitute failure of this test.

4.9 Reliability.

4.9.1 Reliability test conditions. The reliability test shall be conducted in accordance with the requirements of MIL-STD-781 (see 6.2) and as modified by this specification.

a. Temperature. Operating temperature range shall be from $-32^{\circ}C$ to $+52^{\circ}C$.

b. Temperature cycling. Temperature cycling shall be in accordance with figure 4 for the first article reliability demonstration and production group D reliability demonstration. Deviations from temperature specifications are allowed when the chamber door is opened for performance measurements. A thermal survey shall be performed in accordance with MIL-STD-781 prior to initiation of testing.

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c. Vibration. Vibration shall be 2.2 g's, ±10 percent peak acceleration at any non-resonant frequency between 20 and 60 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

d. Duty cycle. Operation of the converter shall be as shown in figure 4.

e. Voltage. Voltage to the converter shall be 18, (+0.5 -0.0) Vdc for the first test cycle, 26 ±1 Vdc for the second cycle and 30, (+0 -1) Vdc for the third cycle. This rotation shall be repeated throughout the test.

4.9.2 Reliability test plan. Five converters shall be selected for first article reliability and three for group D reliability. No single unit's on-time shall be less than half the average operating time of all units on test. Accept/reject criteria shall follow MIL-STD-781, Test Plan IVA.

4.9.3 Performance measurements. Tests as specified below for first article and production reliability shall be made after stabilization of equipment temperature. If during the performance test the value of any specified parameter is not within tolerance, a failure shall be recorded and corrective action taken in accordance with the contract (see 6.2). After appropriate corrective action, testing shall be resumed at the time of the last successful measurement except when pattern failures occur. When pattern failures are detected, 5.7 of MIL-STD-781B shall apply. (A pattern failure is defined as more than one failure, independent and dependent, of components/piece parts for the same defect). Performance tests shall consist of the following:

a. Performance test A. This test shall be performed where indicated in figure 4. It shall consist of measurements as follows:

Regulated dc voltages	3.7.5

b. Performance test B. This test shall be performed once each 24 hours during the cold temperature operation period for odd numbered days and during high temperature for even numbered days throughout the test. It shall consist of measurements as follows:

Reverse voltage protection	3.7.2
Overvoltage protection	3.7.3
400 Hz output	3.7.4
Regulated dc voltage	3.7.5
Overcurrent/short circuit	3.7.6

4.10 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with the requirements of Section 5.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Power Converter, AN/VSG-2, shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Power Converter, AN/VSG-2, shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed Method IC-1.

5.1.1.4.2 Power Converter, AN/VSG-2. Each Power Converter, AN/VSG-2, shall be individually packed Method III as follows: Place each unit within a barrier bag fabricated of material conforming to MIL-B-117, type I, class B, style 2. Closure may be by staples, tape, adhesive or heat seal. Cushion each unit on all surfaces with material conforming to MIL-P-26514, type I, class 1, designed to protect all projections and absorb the shock of impact in handling and transit. Place each cushioned item into a close-fitting box conforming to PPP-B-601, overseas type. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the cover of the box. To assure reusability of the box and to provide easy removal of the cover, fasteners-container closure conforming to MS-24535 shall be utilized. Strapping shall be in accordance with the appendix of the box specification, except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188 (see 6.2).

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. Each Power Converter, AN/VSG-2, packed as specified in 5.1.1, shall require no further packing. The unit container shall serve as the shipping container.

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5.2.1.1 Unitization. Palletization shall be required when: Containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. Each Power Converter, AN/VSG-2, packed as specified in 5.1.1, shall require no further packing. The unit container shall serve as the shipping container, except that the box shall be domestic type.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Power Converter, AN/VSG-2, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or W3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188 (see 6.2).

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129 and as follows:

"REUSABLE PACKAGE
DO NOT DESTROY
DO NOT OPEN PRIOR TO USE"

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The converter is intended for use with the AN/VSG-2 on the M60A3 Tank.

6.2 Ordering data. Procurement documents will specify, as a minimum, the following:

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a. Title, number and date of this specification.
b. That quality conformance inspection lots for group C testing (see 4.4.5) shall correspond to the quantity of converters to be delivered each month.
c. If there is to be a requirement to perform first article tests, a quantity of 5 first article converters are required (see 4.3).
d. Corrective action requirements for group C inspection (see 4.4.5.4).
e. Disposition of first article converters. First article converters shall be returned to satisfactory operation and submitted for inspection and acceptance as part of a quality conformance inspection lot.
f. Level A, level B or commercial preservation and packing (see Section 5).
g. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
h. If the converter is to be subjected to environmental testing as part of the AN/VSG-2 system level test at a sampling rate equal to or greater than that contained herein, then the environmental tests of this specification may be waived.
i. MIL-STD-810B shall be used for environmental tests of 4.8.
j. MIL-STD-781B shall be used for reliability tests of 4.9.
k. The contract shall specify when first article reliability demonstration, production reliability qualification group D tests will be performed and the quantity to be selected for the respective tests, if different from that specified herein. MIL-STD-781B shall be used when conducting the first article group D test.

6.3 Definitions. For the purpose of this specification, the following definitions apply.

6.3.1 Damage. Damage is defined as degradation in specified performance, breakage, loosening, shift, failure of any finish, hardware, connection or component, evidence of condensation or moisture with the system, and deterioration in quality, or failure to operate in accordance with the requirements of this specification.

6.3.2 "g". g is defined as an acceleration or deceleration of 9.8 meters per second per second.

6.3.3 Ambient conditions. Ambient conditions are defined as follows:

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Temperature	23°C, ±10°C (73°F, ±18°F)
Relative humidity	50 percent, ±30 percent
Atmospheric pressure	725, +50 mm of mercury -75 mm of mercury (28.5, +2.0 inches of mercury -3.0 inches of mercury)

CUSTODIAN:
Army - ER

PREPARING ACTIVITY:
Army - ER

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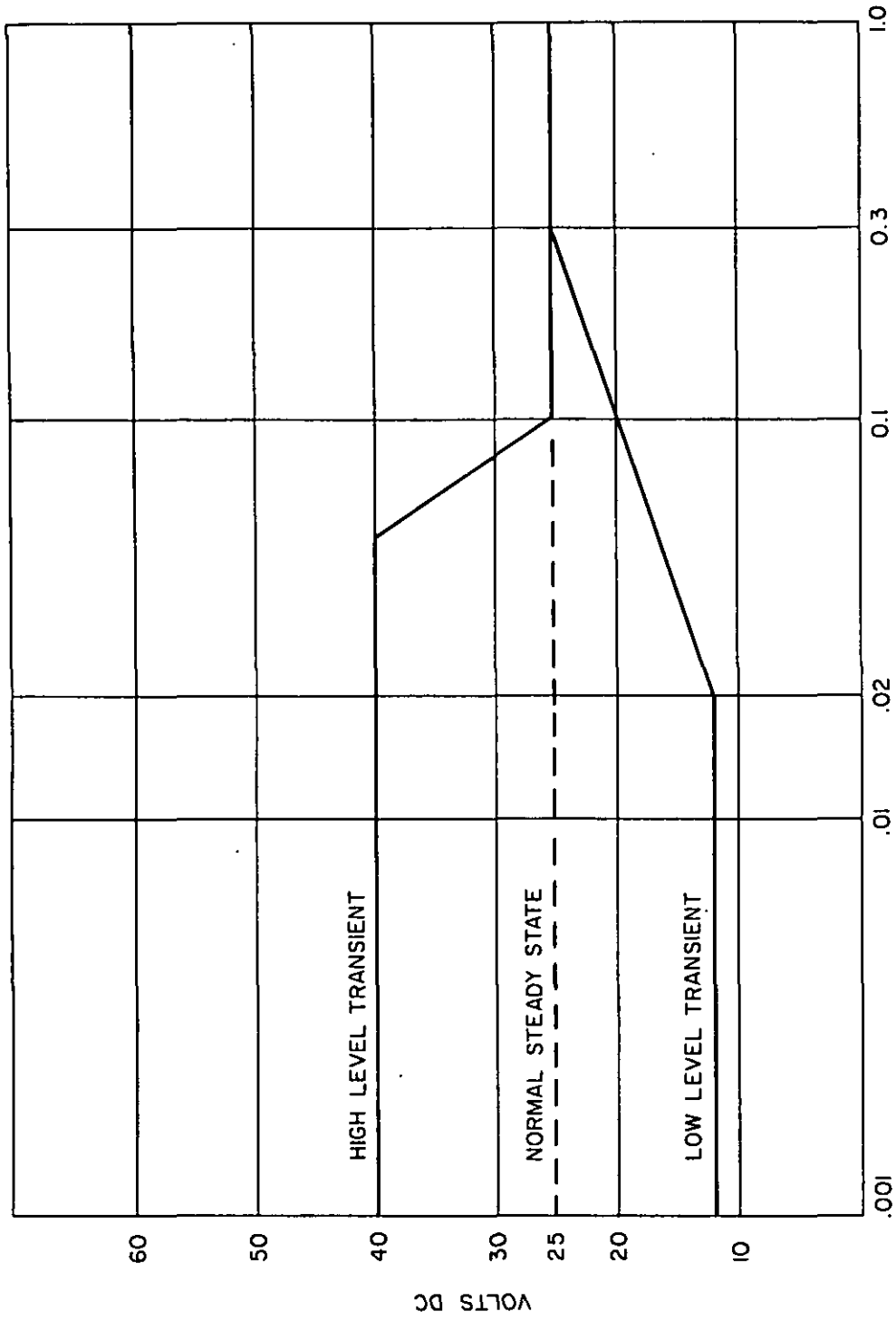


FIGURE 1. TRANSIENT VOLTAGE LEVELS.

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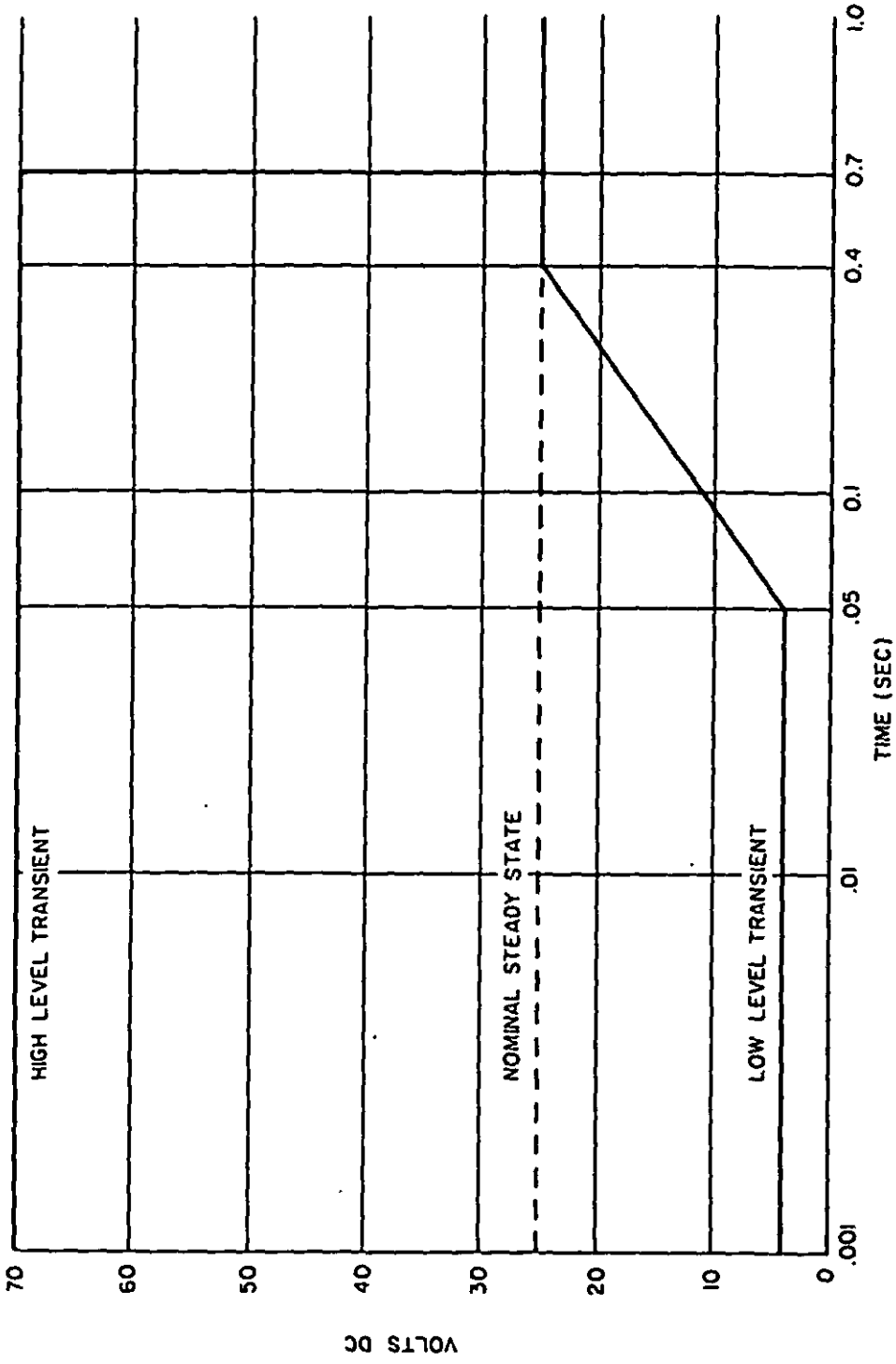


FIGURE 2. SEVERE TRANSIENT VOLTAGE LEVELS.

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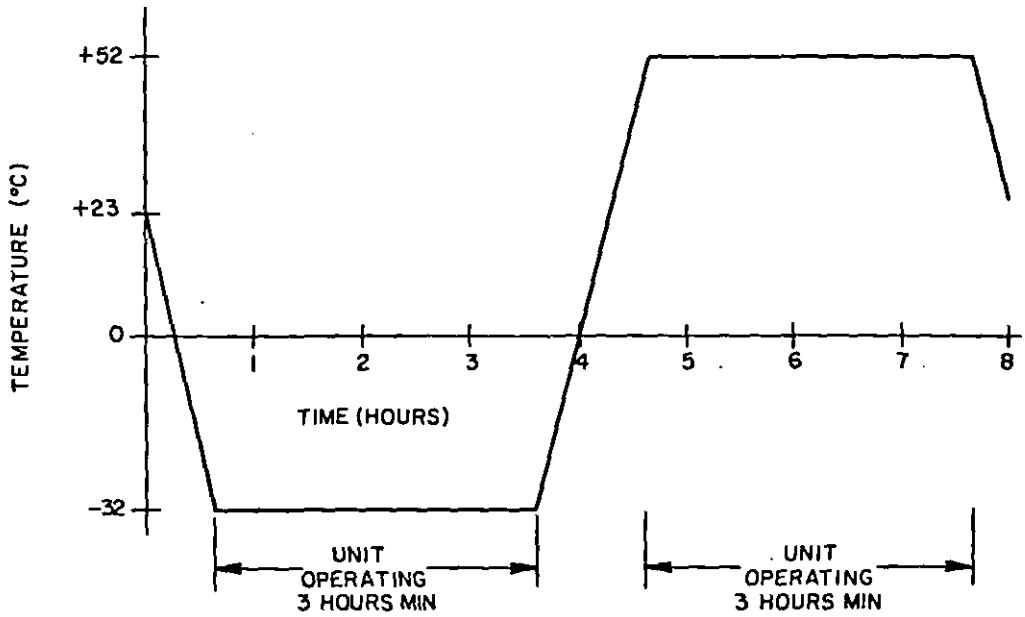
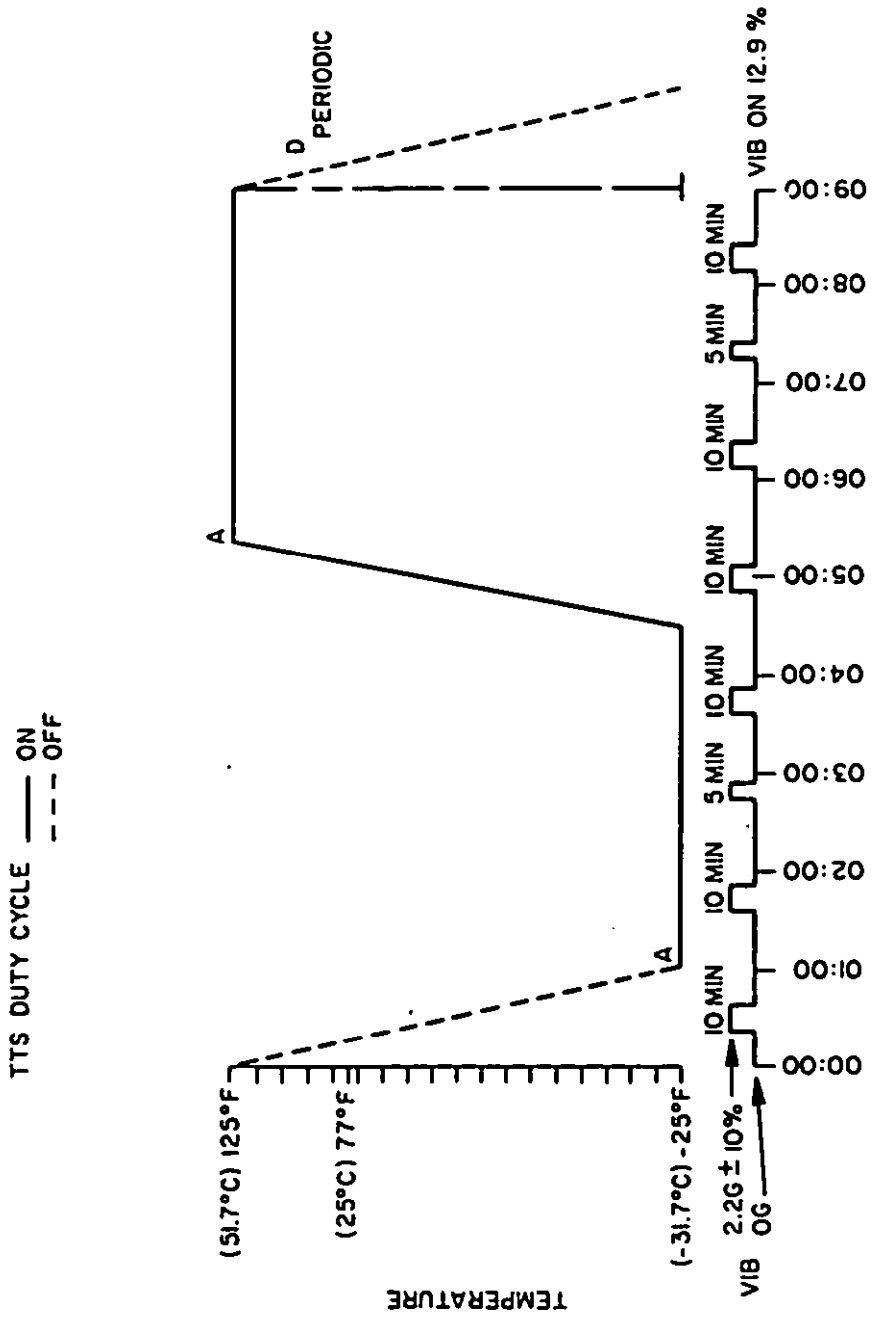


FIGURE 3. ONE PRE-CONDITIONING CYCLE.

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NOTE : CYCLE MAY START OR END AT ROOM AMBIENT, AS REQUIRED.

FIGURE 4. RELIABILITY TEST CYCLE.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-49378(ER)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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